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(%i2) /* N = 4 * m + 1 */
declare(m, integer)$;
assume(m > 0)$;

(%i4) /* Error model: I_true = I_m + C / m^6 */
lm: I(m) + C / m^6;
lm_1: subst(m = m + 1, lm);
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$$lm \quad I(m) + \frac{C}{m^6}$$

$$lm_1 \quad I(m+1) + \frac{C}{(m+1)^6}$$

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(%i5) /* lm = lm_1. Find C constant */
C_: linsolve(lm = lm_1, C);
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$$C_ \quad [C = (m^{12} + 6m^{11} + 15m^{10} + 20m^9 + 15m^8 + 6m^7 + m^6) I(m+1) + (-m^{12} - 6m^{11} - 15m^{10} - 20m^9 - 15m^8 - 6m^7 - m^6) I(m) /$$

$$]$$

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(%i6) /* Estimated absolute error of I(m) */
err_lm: abs(factor(subst(C_, lm - I(m))));
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$$err_lm \quad \frac{(m+1)^6 |I(m+1) - I(m)|}{(2m+1)(m^2+m+1)(3m^2+3m+1)}$$

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(%i10) /* Some formulas for m small */
subst(m = 1, err_lm); /* N = 5 */
subst(m = 2, err_lm); /* N = 9 */
subst(m = 3, err_lm); /* N = 13 */
subst(m = 4, err_lm); /* N = 17 */
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$$(\%o7) \quad \frac{64 |I(2) - I(1)|}{63}$$

$$(\%o8) \quad \frac{729 |I(3) - I(2)|}{665}$$

$$(\%o9) \quad \frac{4096 |I(4) - I(3)|}{3367}$$

$$(\%o10) \quad \frac{15625 |I(5) - I(4)|}{11529}$$